Neural Quantum Embeddings with Multiple Control Variables

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***Abstract*—We investigate the application of Neural Quantum Embedding and Quantum Neural Networks for sentiment analysis using the Stanford Sentiment Treebank dataset. We adapt the Neural Quantum Embedding framework, originally worked with image classification, to textual data by employing quantum embedding techniques that maximize trace distance for better data separability. Additionally, we incorporate advanced quantum feature maps and preprocessing techniques from recent quantum machine learning studies to enhance classification performance. Our experimental results demonstrate that the quantum models catch up with the classical baselines in sentiment classification accuracy, particularly in noisy intermediate-scale quantum settings. This work highlights the feasibility and potential of quantum-assisted sentiment analysis.**

***Index Terms*—Quantum Neural Networks, Neural Quantum Embedding, Quantum Feature Maps, Sentiment Analysis**

I. Introduction

Quantum computing combined with machine learning is evolving rapidly and shows promising synergies in applying quantum-enhanced feature spaces for classical tasks. While classical machine learning models demonstrated impressive performance in natural language processing (NLP), they are inherently constrained by classical feature spaces that may limit their ability to capture complex correlations in high-dimensional data. Quantum neural networks (QNNs) [1] have the potential to enhance these models by encoding data into high-dimensional quantum feature spaces [2]. The recent development of neural quantum embedding (NQE) [3] provides a novel paradigm for integrating quantum representations with deep learning, potentially capturing correlations that may be challenging for classical models in classification tasks.

We consider the image classification application using NQE [4], [5] and extend this approach to natural language tasks. In classical machine learning, multiple models have set new benchmarks in natural language understanding by effectively capturing contextual representations of text. However, classical feature representations are often face dimensionality limits which hinders efficient separation of complex data distributions. Quantum embeddings address this limitation by mapping data to Hilbert space representations, where quantum entanglement and superposition enable non-classical correlations that are difficult to capture with classical models alone. This motivates our investigation into whether quantum-augmented embeddings can enhance NLP tasks such as sentiment classification [6].

We study an application of NQE to the sentiment analysis task using the Stanford Sentiment Treebank (SST) dataset [7]. We introduce a hybrid approach using pretrained deep learning models (BERT [8] and T5 [9]) to extract contextual features and encoded into quantum embeddings. We explore multiple ansatz architectures [10], [11] in the quantum embedding layer and analyze their effects on classification accuracy and robustness. By bridging quantum machine learning and NLP, our research provides new insights into the feasibility of hybrid quantum-classical models and contributes to the broader discussion on the practical utility of quantum embeddings in real-world machine learning applications.

II. Background

A. Neural Quantum Embedding

NQE was introduced as an alternative approach that optimizes quantum embeddings using classical deep learning techniques [2]–[5]. Unlike conventional quantum embeddings that directly parameterize a quantum circuit, NQE employs a neural network to enhance the distinguishability of quantum states before encoding them into quantum circuits. This hybrid approach reduces the depth of quantum circuits required for effective feature encoding and improves the gradient-based optimization during training by mitigating barren plateaus.

B. Quantum Feature Maps

Quantum feature map uses the properties of quantum Hilbert space as an advantage while transforming classical data into quantum states to improve data separability. The success of a quantum model depends greatly on its feature map quality because it defines how effectively the quantum embedding retains and enhances meaningful patterns in input data. Kernel-enhancing feature maps are designed to maximize the expressivity and separability of data in the quantum feature space while ensuring that the inner product between quantum states produces a meaningful similarity measure. We apply various feature maps to measure the representational power of the quantum embedding and to strengthen classification performance.

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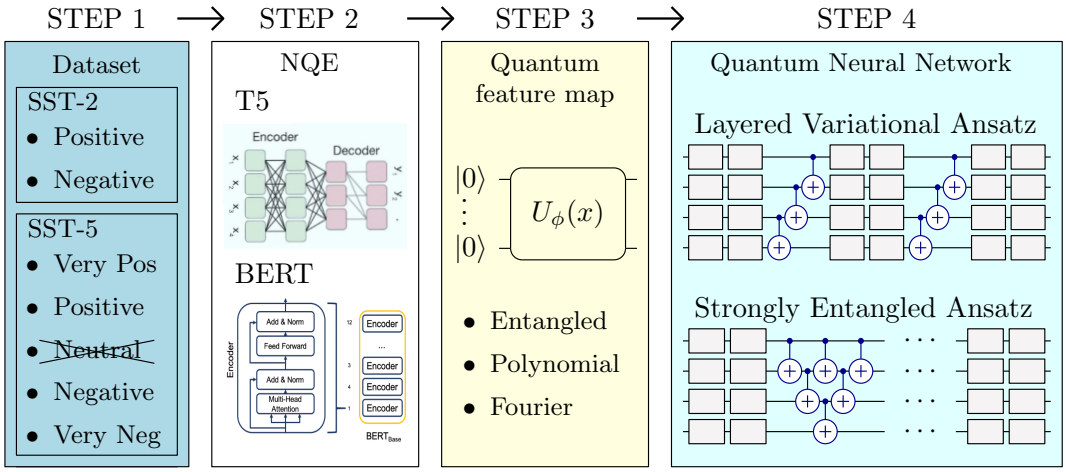


Fig. 1. An overview of our experiment steps. We preprocess two datasets in step 1 and fine-tune these data on two pretrained models (T5 and Bert) in step 2. Then, we combine different feature map types and ansatz types in steps 3 and 4 to configure the quantum circuit. We consider various combinations of these components to verify the impact on how the quantum model learns and generalizes.

C. Quantum Neural Networks

QNNs are a type of parameterized quantum circuit that processes data encoded in quantum states by applying unitary transformations and quantum measurements for learning and classification. The core design of a QNN lies in its ansatz that defines how quantum gates manipulate the input quantum state. The choice of ansatz significantly impacts the expressivity, trainability, and robustness of QNN [12]–[14] and makes it a crucial design factor in quantum-enhanced learning tasks. For our experiments on the SST dataset, we systematically evaluate how different ansatz architectures influence classification accuracy, robustness, and computational efficiency. By comparing various ansatz architectures, we aim to provide empirical insights into the trade-offs between trainability and expressivity in quantum-enhanced sentiment classification.

D. Stanford Sentiment Treebank

The SST dataset [7] is a widely used benchmark for sentiment classification consisting of movie review sentences labeled with sentiment scores ranging from strongly negative to strongly positive. We evaluate the quantum neural embedding framework using two different classification settings. The first is a binary classification task directly from SST-2, where the model distinguishes between positive and negative sentiments. The second task is derived from SST-5 to a four-class setting for a fine-grained sentiment analysis, where we use the class of very negative, negative, positive, and very positive.

III. Methodology

A. Data Preprocessing and Classical-to-Quantum Mapping

We first extract vectorized sentence embeddings using pretrained BERT and T5 models. Since quantum devices have constraints on the number of qubits and allowable input dimensions, we apply dimensionality reduction using a fine-tuned, fully connected layer to map the embeddings into a lower-dimensional space suitable for quantum encoding. Then, the fully connected layer is trained with stochastic gradient descent using a learning rate of 0.01 and optimized with a fidelity loss function to ensure that different classification classes are mapped to well-separated vector spaces. This process is conducted for 1,000 training steps for convergence.

In the next step, the transformed embeddings are mapped into quantum states using feature maps. Quantum states are structured in a high-dimensional Hilbert space to optimize class separability. These feature maps enhance the distinguishability of different sentiment classes by encoding classical features into distinct quantum state representations. This ensures that QNNs operate on well-structured and well-separated embeddings and ultimately improve the classification performance.

B. Feature Maps

Entangled feature maps [15], [16] use multi-qubit interactions to create complex correlations. We use a well-known example, the ZZ feature map, which applies parameterized single-qubit rotations followed by controlled-phase (CZ) entangling gates. This allows the feature map to apply a data-dependent phase rotation to each qubit and a controlled-phase entanglement between qubits i and j, introducing correlations between the qubits. The feature space can achieve increased expressibility including the ability to approximate Haar-random states [17].

Higher-order polynomial feature maps [18] extend traditional polynomial kernel transformations by encoding multivariable interactions into a quantum state. We utilize the Pauli feature map, which applies parameterized single-qubit rotations followed by multi-qubit entangling gates to capture polynomial relationships. This allows the feature map to apply data-dependent phase rotations and higher-degree interaction

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terms between qubits to ensure that polynomial features are embedded.

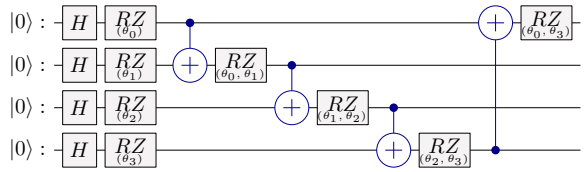


Fig. 2. An example of Higher-order feature map.

Fourier-based quantum feature maps [19] transform the input space into a structured frequency-based representation. We employ a frequency-encoded quantum feature map, where data-dependent phase rotations are applied in repeated layers to simulate Fourier basis functions. This allows the feature map to encode classical data using Fourier frequency components, ensuring that periodic patterns and smoothly varying relationships are well-represented. By mapping input features into a Fourier-expanded quantum state, the mapped feature vectors can capture continuous variations and complex signal structures. This enables the quantum model to recognize periodic trends and long-range dependencies efficiently.

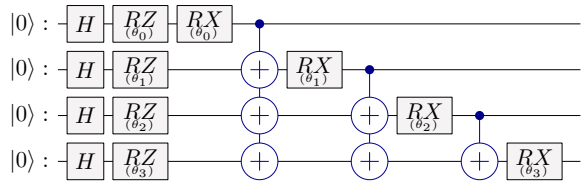


Fig. 3. An example of Fourier-based feature map.

C. Ansatz Architecture

1) Layered Variational Ansatz: The layered variational ansatz consists of alternating layers of parameterized single-qubit rotations (RY and RZ) and nearest-neighbor entangling CNOT gates in a circular pattern. Each layer applies independent trainable parameters to each qubit, allowing for flexible transformations of the quantum state. The entanglement structure ensures that information is efficiently shared across qubits while maintaining a relatively shallow circuit depth, making it well-suited for near-term quantum devices.

2) Strongly Entangled Ansatz: The strongly entangled ansatz introduces full entanglement by applying controlled operations between all qubit pairs. In addition to single-qubit parameterized rotations (RY and RZ), this ansatz employs a ladder of CNOT gates so that each qubit is highly correlated with every qubit in the system. This structure makes it ideal for complex quantum transformations and allow high expressivity where intricate multi-qubit interactions are necessary. However, the increased depth and number of entangling gates can lead to optimization challenges and are prone to errors. This ansatz is great at capturing higher-order correlations.

IV. Experiments and Results

A. Experimental Setup

We conduct a comparative analysis focusing on four key control variables for evaluating the effectiveness of NQE in sentiment classification: NQE models, feature map selection, QNN architecture, and dataset complexity. We assess their impact on classification accuracy and training efficiency in a quantum-enhanced NLP setting by varying these components. All experiments are performed on two execution environments: the Pennylane simulator [20], and IBM Yonsei quantum computer that has the Eagle r3 processor[1] from IBM. We use an average from five runs from the simulator results and a single execution result from the IBM Yonsei due to hardware constraints. The quantum circuits consist of four qubits when paired with the SST-2 dataset, and five qubits when paired with the SST-5 dataset.

B. Experimental Results

Table I presents the accuracy results across different models, feature maps, and quantum architectures evaluated on a Pennylane simulator (left) and IBM Yonsei hardware (right). The results indicate that while quantum embeddings achieve comparable performance to classical baselines, their effectiveness depends on the choice of feature map and dataset complexity.

C. Analysis

We analyze the effects of NQE models (BERT vs T5), feature maps (ZZ, Higher-order Polynomial, Fourier), ansatz (Layered Variational vs Strongly Entangled), and dataset complexity (SST-2 vs SST-5) on sentiment classification performance. While the results align with expectations from prior quantum machine learning (QML) studies [21]–[23], they also highlight challenges and limitations in fine-grained sentiment classification using QNNs. Given the growing research in QML, our findings should be interpreted in the context of recent works that explore when classical surrogate models can replace QML approaches and under what conditions QML can offer advantages over classical methods.

1) NQE models: BERT consistently outperforms T5 in SST-2, especially when paired with the ZZ feature map and layered variational ansatz. T5 follows closely behind, with a peak SST-2 accuracy of 89.2% (simulator) and 78.6% (IBM Yonsei). This supports that BERT's embeddings provide stronger sentiment separability, making it more effective for binary sentiment classification. However, in SST-5, the performance gap between BERT and T5 is much smaller. The best accuracy for BERT and T5 (50.4% and 50.1% on the simulator, 38.0% and 38.0% on IBM Yonsei) is nearly identical when using the higher-order polynomial feature map with a strongly entangled ansatz. This suggests that fine-grained sentiment classification is equally challenging for both models, potentially due to the difficulty in mapping nuanced sentiment differences into the quantum feature space.

[1]https://docs.quantum.ibm.com/guides/processor-types

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TABLE I
ACCURACY COMPARISON AMONG DIFFERENT METHODS. FOR EACH DATASET, WE REPORT AN AVERAGE OF 5 RUNS ON THE PENNYLANE SIMULATOR AND A ONE-SHOT EXECUTION ON A REAL QUANTUM COMPUTER.

NQE Models	Feature Maps	QNN Types	SST-2 (%)	SST-5 (%)
BERT	ZZ Feature Map	Layered Variational Ansatz	90.0 / 79.5	50.4 / 37.7
	Higher-order polynomial Feature Map		87.4 / 75.9	49.4 / 38.0
	Fourier Feature Map		85.3 / 72.4	47.0 / 34.0
	ZZ Feature Map	Strongly Entangled Ansatz	86.7 / 74.4	49.8 / 37.1
	Higher-order polynomial Feature Map		85.2 / 72.6	50.2 / 35.3
	Fourier Feature Map		83.1 / 72.1	45.1 / 31.2
T5	ZZ Feature Map	Layered Variational Ansatz	89.2 / 78.6	49.9 / 38.0
	Higher-order polynomial Feature Map		87.0 / 76.6	50.1 / 37.6
	Fourier Feature Map		85.2 / 76.3	46.0 / 34.5
	ZZ Feature Map	Strongly Entangled Ansatz	87.0 / 74.6	48.6 / 37.6
	Higher-order polynomial Feature Map		84.9 / 73.4	48.4 / 37.1
	Fourier Feature Map		81.8 / 70.7	45.5 / 31.2

2) Feature maps: The ZZ feature Map performs best overall, particularly in SST-2, where it achieves the highest accuracy in both the simulator and IBM Yonsei. This indicates that entanglement-based encoding enhances feature separability, making it effective for distinguishing sentiment classes in binary classification. The higher-order polynomial feature map follows closely behind, performing slightly worse in SST-2 but excelling in SST-5, where it reaches the highest accuracy (50.2% on the simulator, 38.0% on IBM Yonsei for BERT). This suggests that higher-order polynomial interactions help capture more complex decision boundaries, making them more useful for multi-class sentiment classification. The fourier feature map consistently underperforms, particularly in SST-5, where its accuracy drops to 47.0% (simulator) and 34.0% (IBM Yonsei). This indicating that Fourier-based periodic encoding struggles to separate sentiment categories effectively.

3) Ansatz: The layered variational ansatz generally performs better for SST-2, achieving the highest overall accuracy (90.0% in the simulator, 79.5% on IBM Yonsei) when paired with the ZZ feature map. This suggests that a shallower ansatz with structured entanglement is sufficient for binary classification, providing a balance between expressivity and trainability. In contrast, the Strongly Entangled ansatz performs better in SST-5, particularly when combined with the higher-order polynomial feature map, where it achieves the best fine-grained classification accuracy (50.2% on the simulator, 38.0% on IBM Yonsei). This indicates that deep entanglement improves the model's ability to capture subtle variations in sentiment, which is crucial for handling fine-grained sentiment distinctions. However, this also introduces potential optimization challenges or overfitting issues in simpler binary classification tasks, which are one of the well-known difficulties in training deep variational quantum circuits [24], [25].

4) SST-2 and SST-5 datasets: Across all configurations, SST-2 accuracy is consistently higher than SST-5, confirming that binary sentiment classification is feasible for quantum models. The best SST-2 accuracy reaches 90.0% (simulator) and 79.5 (IBM Yonsei), whereas the best SST-5 accuracy remains below 50.2% (simulator) and 38.0% (IBM Yonsei). This significant drop suggests that quantum feature encoding struggles with fine-grained sentiment classification, where subtle differences between sentiment categories are more difficult to separate. The performance gap between the simulator and hardware is also more pronounced in SST-5, with accuracy dropping by 10%–15% when transitioning from simulation to real hardware, likely due to noise and decoherence effects in quantum hardware. This highlights the necessity for improved feature encoding, better circuit designs, and noise-mitigation strategies to enhance the effectiveness of QNNs in multi-class sentiment classification.

V. CONCLUSIONS AND FUTURE WORK

We have explored the application of NQE for sentiment analysis using quantum neural networks. Our empirical results have shown that quantum embeddings can provide an alternative representation for textual data and have achieved comparable classification performance with classical baselines under specific configurations. However, we have also noticed that the observed improvements have varied depending on the feature map, ansatz selection, and dataset complexity.

While quantum models show competitive accuracy in some cases, our findings show that their advantage over classical methods remains inconclusive. Instead, these findings have highlighted the potential of quantum embeddings for specific learning tasks and have emphasized the need for further investigation into their scalability, expressibility, and trainability. Our future research direction is to optimize quantum circuits to mitigate noise effects and explore hybrid quantum-classical architectures that gain the strengths of both paradigms. Another direction is to modify the size of the classical part and the quantum part to find a better ratio between the two methods.

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